

Qatar-2b

R-1071

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_160420 · created 2026-07-30T13:54:40+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457498.90455 +/- 0.00026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.063
Transit depth (Rp/Rs)^2	7.4e-05 +/- 0.011 [%]
Orbital Inclination (inc)	87.22 +/- 1.73
Airmass coefficient 1 (a1)	0.7 +/- 0.48
Airmass coefficient 2 (a2)	0.28 +/- 0.57
Scatter in the residuals of the lightcurve fit is	67.78 %
Best Comparison Star	#1 - [149, 200]
Optimal Aperture	2.88
Optimal Annulus	3.0
Transit Duration (day)	0.0652 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.063 vs 0.16327 (deviation 2.58σ)
Duration fit vs published	0.0652 d vs 0.0767 d (deviation 2.49σ)
Mid-time O-C	0.1 min
Depth SNR	0.0
Residual scatter	67.78 %

Light curve

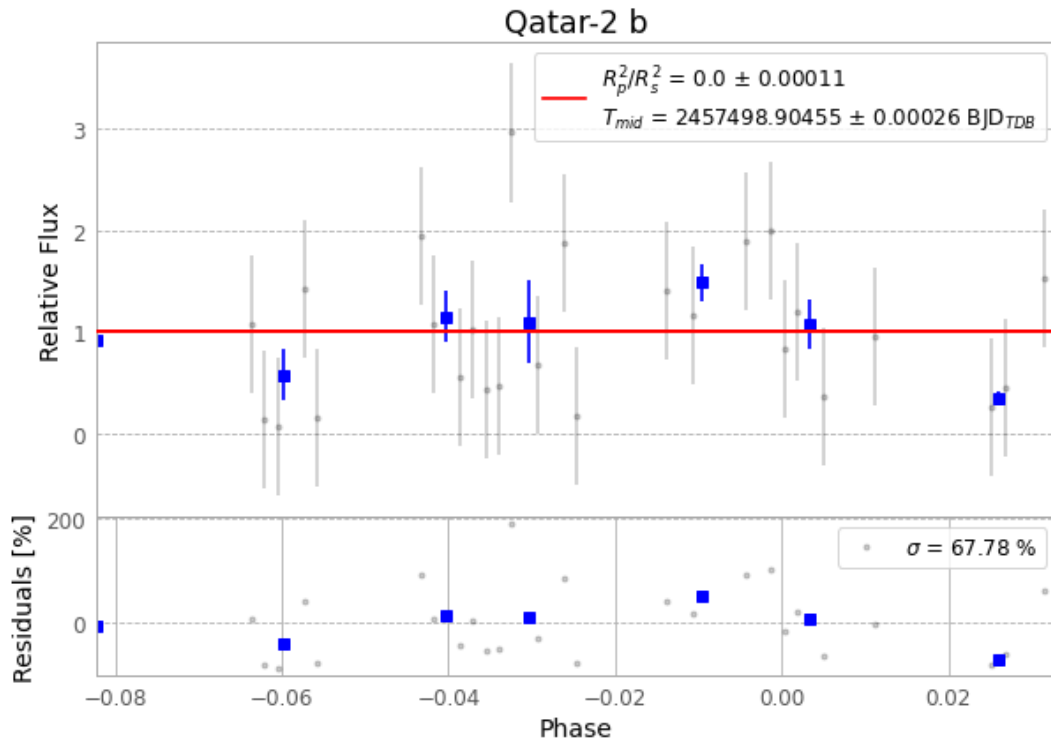


Figure 1 — FinalLightCurve_Qatar-2 b_2016-04-19.png (SHA-256 a156df290e520fe9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [84, 396], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 da0ab9a1fc9859ff...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), Qatar-2160420065412.FITS → Qatar-2160420103912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-160420.log (24acd35a6e18...), FinalLightCurve_Qatar-2 b_2016-04-19.png (a156df290e52...), FinalLightCurve_Qatar-2 b_2016-04-19.csv (d05961c9d6cf...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 13:54Z.